

SOHAILA EMAD

6th of October, Giza, Egypt

sohailaemad85@gmail.com | +20-112-610-4463

github.com/sohaila-emad | linkedin.com/in/sohaila-emad

Portfolio: my-portfolio-two-omega-11.vercel.app

EDUCATION

Cairo University — B.Eng. in Systems & Biomedical Engineering

Oct 2023 – Present (Expected Jul 2027) | Sophomore | CGPA: 3.4 / 4.0

Relevant Coursework: Biomedical Signal Processing, Algorithms, Software Engineering, Machine Learning, Medical Imaging, Data Structures, Differential Equations

TECHNICAL SKILLS

Languages: Python (Advanced), Java, C++, Dart, JavaScript, HTML/CSS

ML / AI: PyTorch, TensorFlow, Keras, U-Net, YOLOv5, XGBoost, SVM/RF, PINNs, Computer Vision

Web & Mobile: React, Flask, Django, DRF, Node.js, Flutter, Firebase

Visualisation: Plotly.js, Chart.js

Tools: Git, PyQt5, SQLite, LaTeX

Biomedical: DSP, Medical Imaging, Beamforming, DICOM, CXR Analysis, Physiological Modeling

AI Tooling: Prompt Engineering, Context Engineering, LLM-assisted development (Claude, GPT-4, Copilot); experienced in designing effective prompts for code generation, debugging, research summarisation, and multi-step reasoning tasks

Soft Skills: Agile, UI/UX Accessibility, Technical Writing

RESEARCH EXPERIENCE

Corneal Curvature Modeling via AI — Cairo University Graduation Project

Feb 2025 – Jun 2025 | PyTorch, Python, PINNs

- Developed a numerical solution for a boundary value ODE modeling corneal curvature using the Method of Lines.
- Implemented a Physics-Informed Neural Network (PINN) approach using PyTorch.
- Compared computational efficiency and accuracy between numerical and ML methods.

TECHNICAL PROJECTS

Signal Viewer: Multi-Modal Analysis Platform

React, Flask, Chart.js

- Built a browser-based biomedical & acoustic signal analysis platform processing multi-channel ECG/EEG, microbiome data, and Doppler audio.
- Integrated fully in-browser AI inference for ECGNet classification and MobileNet for drone audio detection.
- Implemented complex data visualizations: polar plots, recurrence graphs, and STFT spectrograms.

BeamSim: 2D Phased Array Simulator

Flask, React, Python OOP, Signal Processing

- Built a 2D beamforming simulator modeling 5G, Ultrasound (Shepp-Logan phantom), and X-band Radar (PPI) wave propagation.
- Applied physical attenuation, reflection, and interference models within a strictly OOP backend architecture.

Signal Equalizer

React, Flask, NumPy, Plotly.js, DSP

- Developed a real-time audio and ECG signal equalizer with synchronized waveform and spectrogram visualizations.
- Implemented custom frequency band filtering (Generic, Musical, ECG modes) using NumPy FFT and inverse FFT.

FTmixer: Fourier Transform Image Workbench

Django, DRF, React, Image Processing

- Created a dual-mode Fourier Transform workbench to blend images via magnitude/phase and real/imaginary components.
- Engineered an async backend for spatial/frequency transformations (convolution, complex exponential multiply, rotation).

Hybrid AI Pneumonia Detection System

Python, U-Net, YOLOv5, XGBoost, DICOM, CXR Analysis

- Developed a dual-branch AI system integrating classical ML (radiomic features) and deep learning for chest X-ray pneumonia detection.
- Implemented U-Net for lung segmentation; extracted GLCM texture and shape features for XGBoost, Random Forest, and SVM.

3elty – Egyptian Family Health Companion

Flutter, Firebase, SQLite, Agile

- Developed a cross-platform family health app with role-based access control and secure PIN-based authentication.
- Integrated encrypted cloud storage and image compression for digitizing medical records.
- Authored Agile User Stories and Acceptance Criteria for comprehensive feature testing.

Sinus Endoscopy Image Classifier

Python, PyQt5, OpenCV, ResNet18

- Built a GUI application for ENT specialists to classify sinus endoscopy images in real-time; achieved 92% validation accuracy.

ADDITIONAL PROJECTS

3D Anatomy Puzzle Game (Unity3D, Blender, C#) — Educational game with interactive 3D organ models and original assets. | **Medical Brain Image Viewer** (Python, PyQt5) — Synchronized multi-planar MRI viewer with contrast adjustment. | **YOLO Object Tracker** (Python, OpenCV) — Real-time player/ball tracking with movement heatmaps.

LEADERSHIP & ACTIVITIES

IEEE SBME Student Branch

- AI Department Member | 2025 – 2026
- Human Resources Department Member | 2024 – 2025 — Organised recruitment drives attracting 50+ new members; coordinated technical workshops and networking events.